



Alcohol prohibition, social environment and learning: Evidence from India

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Abstract

We study the effects of alcohol prohibition on children's learning outcomes. We exploit a plausibly exogenous change in the availability of alcohol in the Indian state of Bihar, which implemented a state-wide ban on the manufacturing, sale, transport and consumption of alcohol. Using a difference-in-differences framework, we find that the ban reduced alcohol consumption and improved learning outcomes of school-going children in Bihar. The results are plausibly driven by improvements in the child's social environment—lower neighborhood crime and lower domestic violence—induced by the alcohol ban.

Keywords Alcohol · Prohibition · Education · Test Scores · Learning · Bihar

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1 Introduction

Alcohol consumption is a widespread phenomenon in many countries of the world. As of 2019, per capita alcohol consumption (APC hereafter) was as high as 5.43 L of pure alcohol per year among people aged 15 or above worldwide (Ritchie & Roser, 2022). Excessive alcohol consumption is a major public health concern, responsible for over 5% of the global disease burden and 3 million deaths every year (World Health Organization, 2018). Apart from the direct consequences at the individual level such as hospitalizations, morbidity and mortality (Carpenter & Dobkin, 2009; Marcus & Siedler, 2015; Nakaguma & Restrepo, 2018; Barron et al., 2024), it imposes additional costs on societies due to negative externalities like violent crimes, traffic fatalities, suicides, and risky sexual behaviors among other things (Carpenter 2004, 2005a,b, 2007; Biderman et al., 2010; Rossow & Norström, 2012; Baughman et al., 2001; Barron et al., 2024). Moreover, alcohol-induced aggressive behavior is associated with the perpetration of severe physical, emotional and sexual violence on children and fellow family members (Markowitz & Grossman, 1998). Alcoholism may affect child development by disrupting the home environment, inducing family instability and crowding out expenditure on human capital formation (Caces et al., 1999; Gil-González et al., 2006; Collins et al., 2007; Jolex & Kaluwa, 2022).¹ To reduce these alcohol-related harms, countries across the globe have resorted to a variety of policy interventions.²

In this paper, we study the effects of one such policy intervention on children's learning outcomes. The context of our paper is the Indian state of Bihar, which implemented a policy that completely prohibited the manufacturing, sale, transport, and consumption of alcohol. This ban was enacted in April 2016 and was the most stringent alcohol ban in India of the time. Its provisions included heavy fines of INR1,000,000 (approximately US \$16,000, which is about thirteen times India's annual income per capita as of 2014-15), imprisonment of 5 to 10 years and even death penalty in some cases (Chaudhuri et al., 2024). This ban induces a plausibly exogenous variation in the availability of alcohol. Recent research that studies the ban reports that it reduced alcohol consumption, domestic violence and violent crimes in the neighborhood (Dixit & Mukherjee, 2025; Chaudhuri et al., 2024).³ Further, this ban is also shown to have been successful in reallocating household resources away from alcohol consumption to more productive uses such as on education and nutrition of children (Falézan, 2020).

¹ For instance, in China, people who smoke also tend to spend more on alcohol and impact household spending on other productive uses such as human capital formation, farm productivity and financial security (Wang et al., 2006).

² These policies range from regulations on alcohol access—such as restrictions on age, hours of sale and locations of sale—to controls via pricing and taxation. In extreme cases, partial or complete bans on alcohol sales have also been implemented.

³ However, another recent paper by Dar and Sahay (2018) shows that this policy led to an increase in crime in the state.

Our identification strategy exploits the spatial and temporal variation in exposure to the alcohol ban using a difference-in-differences framework. We compare the outcomes of children in Bihar with their counterparts in neighboring Jharkhand before and after the enactment of the policy in April 2016. Using detailed information on children's performance on basic literacy and numeracy tests from the Annual Status of Education Report (ASER hereafter) surveys, we find that the ban increased the reading and math test scores of children in Bihar. In terms of magnitude, the ban increased the reading test score by 0.076 standard deviations and math test scores by 0.102 standard deviations. These results are robust to the inclusion of survey-year fixed effects, birth cohort fixed effects, and a set of controls for individual and household characteristics. Heterogeneity analysis shows that these effects are driven by the relatively richer households in our sample. Further, event studies reveal that the timing of estimated effects closely traces the timing of the implementation of the ban. Our event studies indicate that the effects of the ban are attenuated in 2018, when the government relaxed some of the more stringent provisions under the ban.⁴

Turning to mechanisms, alcohol prohibition can affect test scores via several plausible channels. First, alcohol prohibition can improve the environment outside the household. For instance, alcohol prohibition could reduce crimes in the neighborhood and thus have a positive impact on the test scores of children (Monteiro & Rocha, 2017; Gershenson & Tekin, 2018; Ang, 2020; Michaelsen & Salardi, 2020). Second, a ban on alcohol may lead to improvements in the household environment and parental behavior (especially, the father's behavior in the context of the present study), which in turn can positively affect children's educational outcomes (Koenen et al., 2003; Carrell & Hoekstra, 2010).⁵ Third, a ban on alcohol may shift household expenditures away from temptation goods (such as alcohol) to nutrition and education, which can independently improve test scores (Dahl & Lochner, 2012; Mukherji & Chakraborty, 2024; Krishnatri & Vellakkal, 2026).

In order to shed light on which of these mechanisms drive our results, we use data from the fourth and fifth rounds of the National Family Health Survey (NFHS-4, 2017; NFHS-5, 2021) and the National Crime Records Bureau (NCRB). These datasets allow us to independently verify the effect of this policy on various indicators of the household environment, such as alcohol consumption, intimate partner violence and bargaining power of women inside the household. In line with Dixit and Mukherjee (2025), we find that the ban led to a reduction in alcohol consumption by 0.084 standard deviations and overall intimate partner violence by about 0.5 standard deviations. Further, the ban improved the bargaining power of women within households by 0.256 standard deviations. In line with Falézan (2020), we provide suggestive evidence that the education expenditures increased post-2016. Specifically, we show

⁴ For example, the new regulations provided bail for first-time offenders and prohibited the confiscation of family property in the event that alcohol was found on the property (Chaudhuri et al., 2024).

⁵ Moreover, parents may have more time to spend with their kids which can improve their test scores (Houtenville & Conway, 2008).

that the probability of availing after-class paid private tuition increased post-alcohol ban in Bihar vis-à-vis Jharkhand.⁶

We conduct several robustness checks to further validate our results. First, we show that the improvement in the learning outcomes is driven by children belonging to districts with higher baseline alcohol consumption. Second, we leverage the religious diversity of the population within the state and show that the effects are muted for Muslim-majority regions, which had lower alcohol consumption at the baseline due to religious injunctions on alcohol consumption (Bennett et al., 1998).⁷ Third, we show that our results are robust to redefining the treatment group, alternative identification strategies and alternative measures of the outcome variables.

This paper is related and contributes to at least two distinct strands in the literature. First, an extensive literature studies the link between alcohol and alcohol-related bans on a range of outcomes including homicides and road-accidents (McMillan & Lapham, 2006; Biderman et al., 2010; Green et al., 2014; Martin Bassols & Vall Castello, 2018; Green & Krehic, 2022; Barron et al., 2024), crime and violence (Heaton, 2012; Rossow & Norström, 2012; Dar & Sahay, 2018; Chaudhuri et al., 2024), women's autonomy and fertility (Dixit & Mukherjee, 2025; Majeed & Vellakkal, 2025; Mookerjee et al., 2025). A strand in this literature investigates the link between parental alcohol consumption and children's learning outcomes (see Lowthian (2022) for a review). Existing studies are either correlational or situated in contexts with much higher levels of income per capita (Mangiavacchi & Piccoli, 2018; Raitasalo et al., 2021; Andreas et al., 2022; Pisinger et al., 2023). This paper contributes to the literature by documenting the causal effect of an alcohol ban on learning outcomes (basic literacy and numeracy) in the context of a socio-economically disadvantaged state in a less-developed country. We think that this is an important contribution for the following reasons: Robust correlations (between parental drinking and children's education) documented in the existing literature, while indicative, are not amenable to a causal interpretation. Further, less developed country contexts happen to be very different from contexts with much higher income. For instance, India has achieved near-universal enrollment and instituted de-facto automatic promotion (to higher grades) policies, but learning outcomes in primary and middle school have remained abysmally poor (Muralidharan et al., 2019).⁸ Moreover, the binding constraints in respect of learning outcomes may be very different in less developed countries than in developed countries. The alcohol ban itself may also end up relaxing (or not relaxing) a different set of constraints, given contextual peculiarities like tight household budgets, extremely low levels of maternal autonomy, a high incidence of domestic violence and low state capacity. The effect of parental consumption of alcohol on different outcomes is, therefore, an empirical question. Our paper speaks to the effect of

⁶ Tuition here refers to after school private coaching classes outside of regular school hours. See Section 5.3 for further clarification.

⁷ Unfortunately, we do not have information on a child's religion in our main dataset (i.e., the ASER data). Therefore, we use religious majority at the district level for this analysis.

⁸ For instance, only about 67% of students in grade 8 were able to perfectly complete a math test at the grade-2 level administered by the ASER surveys that we use in this study. Therefore, the margin of adjustment for educational outcomes may be very different in less developed countries (for instance, high school graduation versus basic numeracy).

prohibition on learning outcomes, which, to the best of our knowledge, has not been documented before.

Second, existing studies document high returns to education (Duflo, 2001; Banerjee & Duflo, 2005), and highlight the importance of “quality” of education (actual proficiency) rather than just quantity of education (for example, completed years of schooling) (Hanushek & Woessmann, 2008; Schoellman, 2012). Poor learning outcomes in primary and middle school has been recognized as a problem in the context of many less-developed countries, yet there is a dearth of scalable policies that have been proven to improve learning outcomes in these contexts (Deming, 2022; Glewwe & Muralidharan, 2022). The evidence on the effectiveness of demand-side interventions (such as those that relax the household’s information and liquidity constraints) is mixed (Barham et al., 2013; Loyalka et al., 2013; Benhassine et al., 2015). Supply-side interventions that focus on the provision of learning inputs (like textbooks and uniforms) or unconditional increases in teachers’ salaries have been found to be ineffective (Borkum et al., 2012; Hidalgo et al., 2013; De Ree et al., 2018). Technology-focused interventions have been found to be effective, especially when the technology personalizes content (Muralidharan et al., 2019; Naik et al., 2020), but such interventions are costly and difficult to administer at scale. Existing studies also find that interventions that potentially relax both demand and supply-side constraints such as school meals and a cycling-to-school program are effective in improving access to schooling and learning outcomes (Muralidharan & Prakash, 2017; Chakraborty & Jayaraman, 2019). This paper contributes to the existing literature by documenting that an unlikely candidate, namely, a total ban on alcohol, is effective in improving learning outcomes in the Indian context.

Third, our paper adds to the ongoing debate on the welfare implications of alcohol prohibition policies. Indian states like Bihar are heavily dependent on “sin taxes” on commodities like alcohol for revenue, and the alcohol ban led to a loss of INR 31,417 million (about 12% of the state GDP) to the state of Bihar. This has led some scholars to advise against alcohol bans as a policy recommendation (see the discussion in Ch. 7 on pg223 in Muralidharan (2024)). Our analysis contributes to a series of recent papers that document positive effects (such as lower crime, domestic violence and greater maternal autonomy, education and fertility outcomes) induced by the ban (Chaudhuri et al., 2024; Dixit & Mukherjee, 2025; Falézan, 2020; Chakrabarti et al., 2024; Krishnatri & Vellakkal, 2026; Majeed & Vellakkal, 2025). Along with these existing studies, this paper suggests the need for more nuance in the policy discourse around alcohol bans in the context of less-developed countries. Back-of-the-envelope calculations that we conduct indicate that improvements in learning outcomes alone could recover about one-third of the revenue lost by the state due to the ban. This suggests that when all the benefits (lower crime, domestic violence and better children’s nutrition and learning outcomes) are considered, it is possible that the policy yields net positive welfare gains. While a full-fledged welfare analysis is beyond the scope of this paper, this is, to the best of our knowledge, the first attempt to quantify the gains from the Bihar alcohol ban.

The remainder of this paper is organized as follows: Section 2 provides a brief institutional background. Section 3 describes the data and our empirical strategy. Section 4 presents the results. Section 5 explores potential mechanisms. Section 6

presents a battery of robustness checks. Section 7 presents a back-of-the-envelope cost-benefit analysis of the ban. Section 8 concludes.

2 Institutional background

2.1 Alcohol consumption and the Bihar prohibition act (2016)

Alcohol consumption is a major health risk worldwide and was responsible for about 5.4% of total deaths in India—the context of this paper—in 2016 (World Health Organization, 2025). With APC of 5.7 L of pure alcohol per person, India is one of the top alcohol-consuming countries, and its APC is projected to increase by 2.2 L by 2025, contributing significantly to the overall rise in total APC in South-East Asia, as projected by WHO (World Health Organization, 2018).⁹

India has a federal polity, and the regulation of alcohol is a state subject. In order to curb the excessive consumption of alcohol in the country, a multitude of measures, including minimum legal drinking age (MLDA), taxation and partial or total alcohol prohibitions, have been adopted by different Indian states. Since independence, about 13 complete or partial alcohol bans have been implemented in India. Among these, the Bihar Prohibition and Excise Act (2016), which we focus on in this paper, stands out as the latest and the most stringent act.

Bihar is a state in the northeastern part of India and shares an international boundary with Nepal (see Fig. 1). It is one of the most populous states and ranks amongst the most backward in India, with a low level of socioeconomic development and literacy rate. Bihar had the sixth highest alcohol consumption in the country with an APC of 14.7 L per year (NFHS-4, 2017).¹⁰ During 2005–2016, the number of alcohol shops increased and doubled in the state of Bihar. In the decade preceding the ban, 29% of the men aged 15 years or above reported consuming alcohol. The incidence of alcohol consumption was much lower amongst women, with less than 1% women reporting that they consume alcohol (NFHS-4, 2017).¹¹ The year 2015 witnessed a spate of women-led protests against domestic violence, and protesters clearly indicated alcoholism amongst men as a proximate cause. State elections were due the next year (i.e., 2016), and, at that time, campaigning was on in full swing. A group of women voters approached the incumbent chief minister and chief ministerial candidate, Mr. Nitish Kumar, demanding a complete ban on alcohol. The protesting women argued that in order to reduce the rising incidence of domestic violence, alcohol should be banned. Nitish Kumar, the incumbent, won the election, and within three weeks of taking office, he announced that a complete ban on alcohol would be operational from 1st April 2016, the beginning of the next fiscal year.

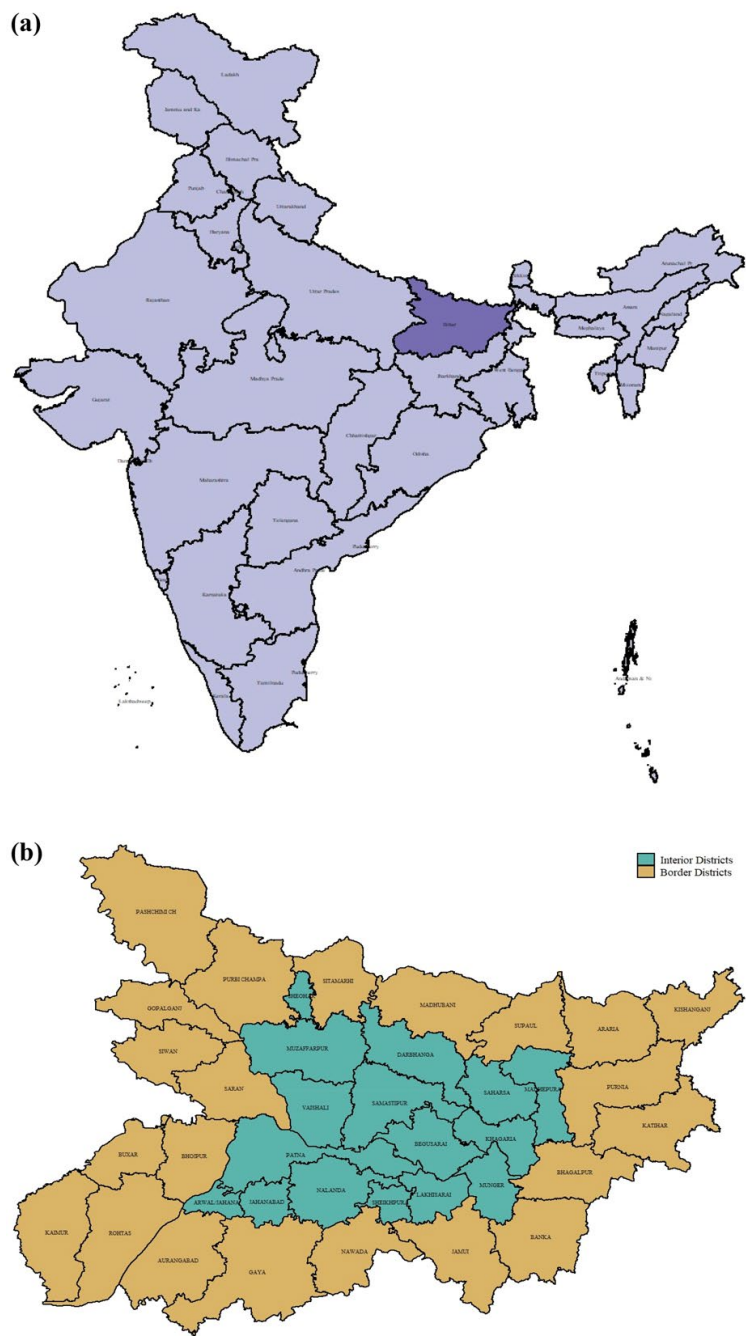
Under this law, the Bihar government prohibited the manufacturing, transport, sale and consumption of alcohol throughout the state with stringent provisions for the

⁹ To the best of our knowledge, at the time of writing this paper, the updated and directly comparable data for 2025 is not available yet.

¹⁰ These levels are more than double the national level APC.

¹¹ These rising figures are almost similar to developing countries like South Africa and Brazil.

Fig. 1 The location and district map of Indian state of Bihar



offenders. However, it was noted that the past alcohol prohibitions failed due to their weak enforcement, black marketing and leakages from non-dry neighboring states. Learning from these failures, the Bihar government implemented a draconian version of the ban with huge fines of INR 1,000,000 (USD \$16,000), 5 to 10 years of imprisonment and even the death penalty in some cases for the offenders (Chaudhuri et al., 2024). A toll-free number was painted on walls to encourage people to report cases of drinking and nuisance to police. Subsequently, over 150,000 people were arrested for violating the ban since April 2016, and INR 50 million (USD \$1 million), worth stock of alcohol was seized by the government (Falézan, 2020). In the year preceding the ban, excise tax revenue from alcohol amounted to INR 31,417 million, accounting for approximately 12% of the total government revenue (Dixit & Mukherjee, 2025). Despite the concerns that the ban would reduce developmental expenditure and affect

long-term growth due to revenue loss, the ban proved successful in reducing violence against women and criminal activities in the state (Chakrabarti et al., 2024; Chaudhuri et al., 2024). However, in 2018, the Bihar government decided to ease several stringent provisions of the ban, including the removal of the property seizure clause if liquor is found in private premises and allowing bail for first-time offenders.

2.2 Learning outcomes in Bihar

Bihar—one of the most impoverished and backward states of India—fares poorly in terms of all the Human Development Indicators.¹² For instance, in 2011, Bihar had the lowest literacy rates among all the states, both for women (50%) and men (78%), which are far below the national averages of 68% for women and 86% for men. Moreover, school attendance among children in Bihar is also very modest. The ASER 2014 report shows that, on average, 52.1% of the enrolled students in grades I–VII/VIII attended schools on the day of the survey in Bihar, whereas in some of the northern states of India, this figure is around 80–90%.¹³ Moreover, the ASER reports show that the learning deficits in Bihar are also very large. For instance, about 78.2% of third graders cannot read a short story equivalent to a second-grade level textbook, and about 87.3% of them cannot even do a simple division.

3 Data and empirical strategy

3.1 The data and descriptive statistics

We use data from several rounds of the Annual Status of Education Report Surveys (ASER, 2019). The primary objective of these surveys is to assess the foundational skills of children in rural India. The ASER is a household-based repeated cross-sectional survey that collects extensive information on children's learning outcomes and their basic demographic and household characteristics. Since 2005, the surveys have been conducted each year in about 580 districts of India (Chakraborty & Jayaraman, 2019),¹⁴ typically collected around November or December (Fagernäs & Pelkonen, 2020).

The ASER surveys collect information on children's enrollment status (for children aged 3–16) and reading and math proficiency of children aged 5–16, irrespective of their enrollment status. The tests are administered in local languages using rigorously developed tools.¹⁵ In the reading test, a child is first asked to read a paragraph (equivalent to that in a grade 1 textbook). If the child can read a paragraph, she is next

¹² These arguments are based on the data from the Sub-national HDI Database of the Global Data Lab (Smits & Permanyer, 2019).

¹³ Appendix Figure D.1 plots the yearly attendance rates for the states of Bihar and Jharkhand against the national average. It shows that the two neighboring states have similarly very low rates of attendance, both in absolute terms as well as relative to the national average.

¹⁴ From 2005 to 2014, the survey was conducted every year. However, after 2014, the survey was conducted every alternate year.

¹⁵ For instance, see [https://asercentre.org/wp-content/uploads/2022/12/English ASER-2018.pdf](https://asercentre.org/wp-content/uploads/2022/12/English%20ASER-2018.pdf).

asked to read a short story (equivalent to a grade 2 textbook). If the child cannot read the paragraph, she is asked to read a word. If she cannot read a word, she is asked to read the alphabet. The child's performance is evaluated at the highest level of mastery. That is, if a child can read a paragraph but not a story, then she is assumed to have mastered lower levels than a paragraph (such as words and alphabet). Thus, the outcomes of the reading test are a categorical variable. Each child is assessed to have achieved one of the following four levels of mastery based on their ability: to read alphabets, words, paragraph and story. Similarly, a child is assessed on four levels of mastery in mathematics/arithmetic. These include her ability to recognize single-digit numbers, double-digit numbers, perform a subtraction problem (two-digits with borrowing—equivalent to grade two level) and a division problem (three-digit by one-digit—equivalent to grade 3–4 level).

The tests administered in the ASER data are foundational, not grade-appropriate. Smaller surveys by individual researchers have documented large gaps in learning at higher grades (Muralidharan et al., 2019). Unfortunately, there is, to the best of our knowledge, no nationally representative dataset that captures information on grade-appropriate learning outcomes. Since we use the ASER data, we are not able to study how the alcohol ban changed grade-appropriate outcomes. This remains a limitation of the current study.

We use the information in the ASER data to construct our test score variables. For reading ability, the score is coded as follows: cannot read (0), read letters (1), read words (2), read a paragraph (3) and read a story (4). For mathematical ability, the score is coded as follows: cannot recognize numbers (0), recognizes single-digit numbers (1), recognizes double-digit numbers (2), can subtract (3), and can divide (4). Following the existing literature (Fagernäs & Pelkonen, 2020; Lahoti & Sahoo, 2020; Krämer et al., 2021), from these categorical variables, we generate *z*-scores by standardizing them with respect to the mean and standard deviation of the control group (i.e., Jharkhand).¹⁶ This allows us to interpret our results as effects in standard deviations.

Since 2009, the survey has consistently collected rich information on a child's background characteristics such as her age, gender, grade, household size, parental schooling status and level of education, and household socioeconomic status, among others. In our empirical analysis, we use data from eight repeated cross-sections from 2009 to 2014, 2016 and 2018.¹⁷ Further, we restrict our sample to children who are in primary school (grades 1–5) or middle school (grades 6–8). We restrict our sample to this group of children for two important reasons: First, since our data on test scores is not grade-appropriate (i.e., all children are given the same basic reading and math tests, irrespective of their grade), we restrict our sample to children up to middle school for whom it is somewhat appropriate. Second, since these children are still very young (mostly less than 14 years of age) and highly unlikely to be consum-

¹⁶ However, as discussed in Section 6, we alternatively standardize these test scores with respect to age of the children.

¹⁷ We restrict our data till 2018 because this is the last survey before COVID-19 pandemic hit India in early 2020.

ing alcohol, the ban should not have a direct impact on their performance via self-drinking behavior.

For our purpose, we restrict our data to the currently enrolled children from the states of Bihar and Jharkhand.¹⁸ We prefer Jharkhand as our control group for two reasons. First, Jharkhand is a neighboring state which was a part of Bihar until 2000, and it comprises 18 southern districts of erstwhile Bihar.¹⁹ Second, Jharkhand has been exclusively used as a counterfactual group for Bihar in several previous studies (Muralidharan & Prakash, 2017; Dar & Sahay, 2018; Chaudhuri et al., 2024). Since the alcohol prohibition policy is a state-wide reform, it is very likely that the policy may not be as effective in the border district of Bihar as it could be in the interior districts of Bihar. This is because of the fact that the movement of goods and services, as well as the population, is not restricted across state borders, unlike international borders (Chaudhuri et al., 2024). Therefore, we further restrict our sample to children belonging only to the interior districts of Bihar (see Fig. 1 for details about interior and border districts of Bihar).

Table 1 presents summary statistics. The average reading and math scores in our sample are about 2, implying that, on average, a child in our sample can read words but cannot anything above that. Similarly, an average child in our sample can recognize double-digit numbers but cannot perform subtraction and division problems. Figure 2 plots the distribution of the reading and math scores. As depicted in Fig. 2a and b, the learning outcomes of children are notably low. 17% of children are not even able to recognize the letters. 19% of children can read letters but struggle to read words. 14% of children can read a paragraph along with letters and words but are not able to read a short story. Only 35% of the children can read at the highest level of reading proficiency. 21% of children can recognize single-digit numbers but struggle with double-digit numbers. 18% children can perform subtraction and recognize both single and double-digit numbers yet face difficulty while performing division. Only 27% of the children tested demonstrate the ability to perform division problems.

The mean and standard deviation of the standardized test scores are not exactly zero and one, respectively, due to the normalization with respect to the control group. About 42% of the children avail after-class paid private tuition. In our sample, about 53% of the children are male. The average child is 10 years old and is currently enrolled in class 4. About 63% of fathers and 38% of mothers are literate. An average household has about 7 members, 23% of them have a toilet facility within the household. The socioeconomic condition of the household is poor, with only about 23% of the houses being made of concrete material.

3.2 Empirical strategy

In this paper, our objective is to investigate the effect of a complete ban on alcohol on children's learning outcomes in the Indian state of Bihar. Our identification strategy exploits spatial (across districts) and temporal (across years) variation in exposure to

¹⁸ However, as a robustness check, we also present results with the full sample of children (irrespective of their enrollment status) sampled by ASER in Section 6.

¹⁹ This makes Jharkhand culturally and socioeconomically more comparable to Bihar than any other state.

Table 1 Summary statistics

	Mean	SD	Min	Max	Observations
<i>Educational Outcomes</i>					
Reading Score	2.301	1.509	0	4	225,938
Standardized Reading Score	0.005	1.032	-1.569	1.167	225,938
Math Score	2.211	1.391	0	4	224,814
Standardized Math Score	0.051	1.043	-1.608	1.394	224,814
Paid Private Tuition	0.419	0.493	0	1	232,466
<i>Child and household level characteristics</i>					
Child Gender (1=Male)	0.528	0.499	0	1	190,024
Age	9.720	2.738	5	16	190,024
School Class	4.095	2.256	1	8	190,024
Mother Literate (1=Yes)	0.376	0.484	0	1	190,024
Father Literate (1=Yes)	0.628	0.483	0	1	190,024
Household Size	6.949	3.244	1	78	190,024
Household Toilet	0.226	0.418	0	1	190,024
<i>Household Type</i>					
— Kutcha (made of mud etc.)	0.522	0.500	0	1	190,024
— Semi-Pucca	0.244	0.430	0	1	190,024
— Pucca (concrete)	0.233	0.423	0	1	190,024

Notes: Authors’ calculations from ASER household level data; the sample is restricted to children from Jharkhand and the interior districts of Bihar; data source is the ASER household surveys from 2009 to 14, 2016 and 2018

alcohol ban in Bihar. It compares the outcomes of children in the districts of Bihar with their counterparts in Jharkhand before and after the enactment of the policy in April 2016 under a difference-in-differences framework. We estimate the following specification:

$$y_{ibdt} = \alpha + \beta (Bihar_d \times Post_t) + \theta X_{ibdt} + \vartheta_b + \delta_d + \tau_t + \epsilon_{ibdt} \tag{1}$$

Here, y_{ibdt} represents the standardized test score of child i born in the year b , who lives in district d and was surveyed in the year t . $Bihar_d$ is our treatment dummy, which takes value 1 for the districts of Bihar and 0 for the districts of Jharkhand. $Post_t$ is an indicator that takes value 1 for the survey years 2016 and 2018 and zero otherwise. X is a vector of control variables that includes a bunch of individual and family characteristics such as the child’s gender, grade, parental education status, family size, socioeconomic status and the availability of a toilet inside the house. ϑ_b , δ_d and τ_t are vectors of birth cohort fixed effects, district fixed effects and survey year fixed effects, respectively.

To capture the pure effect of alcohol ban on test scores, we control for three important dimensions of systematic shocks to our outcomes. First, accounting for the pos-

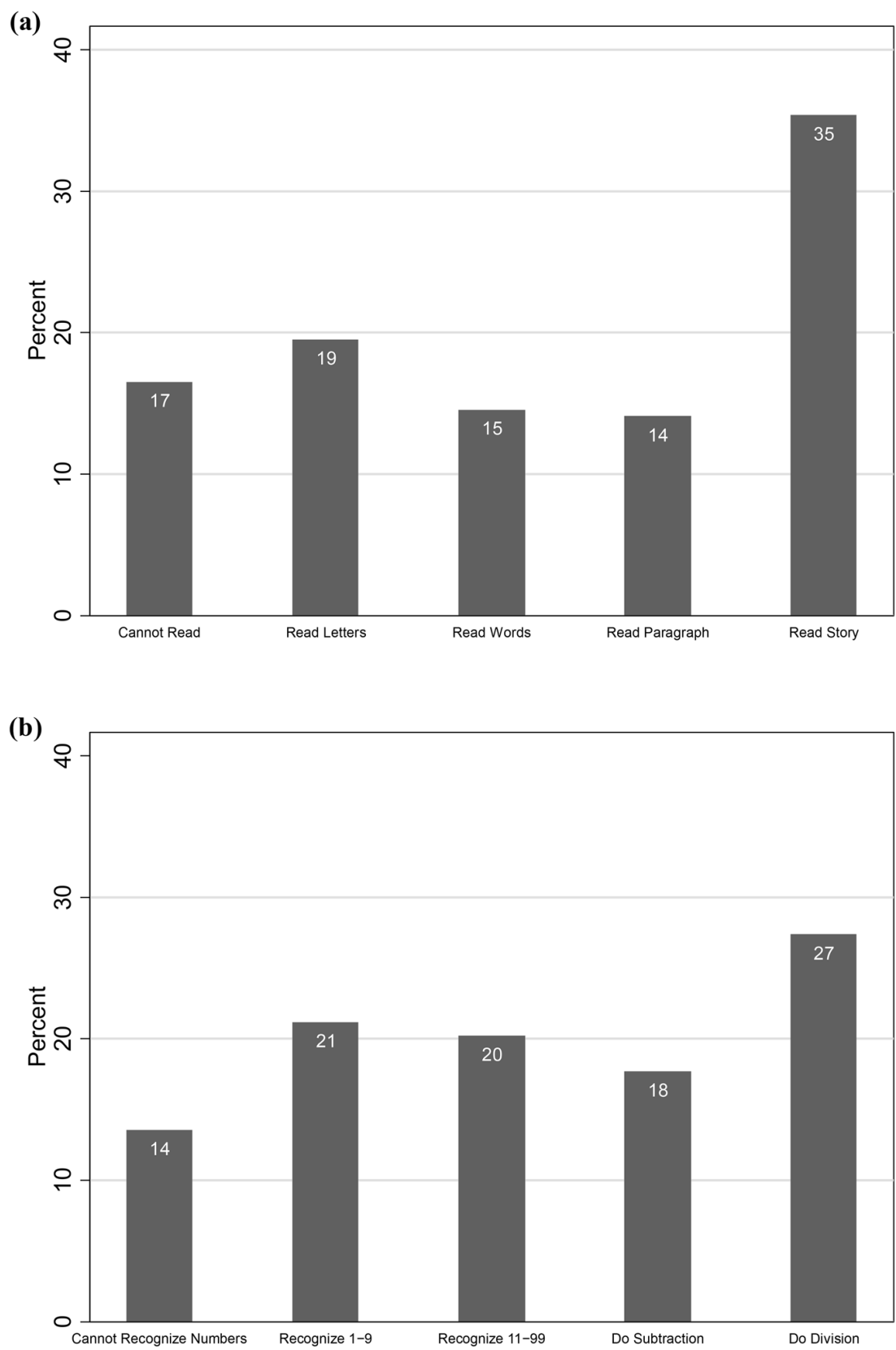


Fig. 2 Distribution of learning outcomes of students in Interior Bihar & Jharkhand

sibility that there could be secular time trends in test scores, we include survey year fixed effects. Second, to account for the fact that older children could do well on these relatively basic language and arithmetic tests, we also include the birth cohort fixed effects. Third, to account for the fact that districts may systematically differ from each other, we also include district fixed effects in our specification.

Moreover, since the alcohol prohibition policy is a state-wide reform, its effectiveness may be limited in Bihar's border districts as compared to Bihar's interior districts. This is because, unlike international borders, the movement of goods and services and people is completely unrestricted across state borders within India (Chaudhuri et al., 2024). The non-dry regions of neighboring states of Bihar could potentially serve as leakage points and may render the ban ineffective in such regions. Therefore, in our main analysis, we restrict our analytical sample only to the interior districts of Bihar and drop all those districts of Bihar that share boundaries with the neighboring states (see panel (b) of Fig. 1). So, our treatment variable takes value 1 for all the interior districts of Bihar and 0 for Jharkhand. In Section 6, we conduct a robustness check showing that our results are robust to redefining the treatment group to include all districts of Bihar.

Identification in a difference-in-differences framework crucially hinges on the assumption of parallel trends, which, in our case, means that the test scores of children in treated (interior) districts of Bihar would have evolved in parallel to their counterparts in Jharkhand. To test the validity of the common trends assumption, we estimate event studies using the following specification:

$$y_{ibdt} = \alpha + \sum_{k=\{2009, \dots, 2018\} \setminus 2014} \gamma_t 1_t \times Bihar_d + \theta X_{ibdt} + \vartheta_b + \delta_d + \tau_t + \epsilon_{ibdt} \quad (2)$$

Here, 1_t is a dummy and takes the value 1 for all observations belonging to year t and zero otherwise. The rest of the specification is the same as Eq. (1). The coefficients γ_t 's in this specification estimate the gaps in the test scores of children in the interior districts of Bihar and their counterparts in Jharkhand every year with respect to the base year 2014. We expect these gaps to be very close to zero and statistically insignificant prior to the enactment of the policy in 2016. This will lend support to our identification assumption of parallel trends.

Recent advances in the literature around difference-in-differences have raised concerns that event studies may lack statistical power to detect violations of the parallel trends assumption. Therefore, we follow Rambachan and Roth (2023) and check the sensitivity of our event study estimates to potential violations of the parallel trends. We present the results from this exercise as a robustness check in Section 6.

4 Results

Table 2 presents results from estimating specification (1) with reading score as the outcome. The reading scores have been standardized with respect to the control group mean and standard deviation. Therefore, the estimates in the table can be interpreted as standard deviations. Column 1 shows the results from the most basic version of

Table 2 Effect of alcohol ban on the reading score of children in Interior Bihar

	Standardized Reading Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior $\times$ Post	0.111*** (0.029)	0.105*** (0.029)	0.109*** (0.028)	0.074** (0.030)	0.076*** (0.028)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	225,938	225,938	225,938	225,668	190,024
R-squared	0.001	0.021	0.027	0.360	0.471

Notes: The dependent variable is the reading score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child’s age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009–14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Eq. (1), which does not include any fixed effects and controls.²⁰ The results indicate that the alcohol ban led to an increase in the reading ability of children in Bihar vis-à-vis their counterparts in neighboring Jharkhand. In terms of magnitude, the policy leads to an increase of 0.11 standard deviations in reading score. This estimated coefficient is statistically significant at the 1% level. The positive effect of the alcohol ban on reading proficiency is robust to the inclusion of district fixed effects (Column 2), survey year fixed effects (Column 3) and birth cohort fixed effects (Column 4). Besides, these effects are robust to the inclusion of a bunch of individual and household characteristics such as the child’s gender, grade, parental education status, family size, socioeconomic status and the availability of toilet inside the house. Overall, using our preferred specification, we find an increase of 0.076 standard deviations in the reading ability of school-going children in Bihar after the alcohol ban was put in place.

Next, Table 3 presents the results for math score, which have been standardized like reading score, so that the coefficients can be interpreted as standard deviations. Column 1 shows the results from estimating the most basic version of the Eq. (1). It does not include any fixed effects and controls for individual and household characteristics. The results indicate that the alcohol ban lead to an increase in the math score for children in Bihar. In terms of magnitude, the math score improved by about 0.145 standard deviations and the effect is statistically significant at the 1% level. In columns 2–5, we estimate the stricter versions of the estimating Eq. (1) and include district fixed effects (Column 2), survey year fixed effects (Column 3), birth cohort fixed effects (Column 4) and a bunch of individual and household characteristics (Column 5). The results remain stable and retain statistical significance. Overall,

²⁰ For columns without district or year fixed effects, we have controlled for the treatment and the post dummy separately in the regression. However, for the reason of brevity, we have not reported them in the Table.

Table 3 Effect of alcohol ban on the math score of children in Interior Bihar

	Standardized Math Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior × Post	0.145*** (0.032)	0.142*** (0.032)	0.144*** (0.031)	0.109*** (0.032)	0.102*** (0.029)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	224,814	224,814	224,814	224,547	189,232
R-squared	0.005	0.025	0.035	0.365	0.478

Notes: The dependent variable is the math score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child’s age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

using our preferred specification, we find an increase of 0.102 standard deviations in the math ability of school-going children in Bihar post the alcohol prohibition act vis-à-vis their counterparts in neighboring Jharkhand.

Overall, the state-wide alcohol ban in Bihar improved the reading and math ability of school-going children by 0.076 and 0.102 standard deviations, respectively. To put our estimates into context, these effects are equivalent to about half the impact of a child’s exposure to mid-day meals throughout primary school on reading scores and slightly exceed their impact on math scores, as documented for India by Chakraborty and Jayaraman (2019).²¹ To further contextualize these estimates, the median effect size for quasi-experimental education interventions in developing countries is 0.05 standard deviations for both reading and math scores, and the 60th percentile of the effect size distribution is 0.08 standard deviations for reading and 0.09 standard deviations for math scores (Evans & Yuan, 2022). Therefore, compared to these benchmark effect sizes, the effects of the alcohol ban on test scores are sizable.

4.1 Identification

Identification in a difference-in-differences framework crucially hinges on the assumption of parallel trends. In our case, this requires that the test scores of children in Bihar should have evolved in parallel to their counterparts in Jharkhand in the absence of the Alcohol Prohibition Act of 2016. We test the validity of the identifying assumption using an event study design. Specifically, we present results from estimating Eq. (2), separately for the reading and math test scores. Figure 3 presents the results. Panels (a) and (b) show the event study plots for the reading and math

²¹ Chakraborty and Jayaraman (2019) show that providing midday meal to a child throughout her primary schooling improves her reading and math ability by 0.17 and 0.09 standard deviations, respectively, compared to a child with less than one year of exposure.

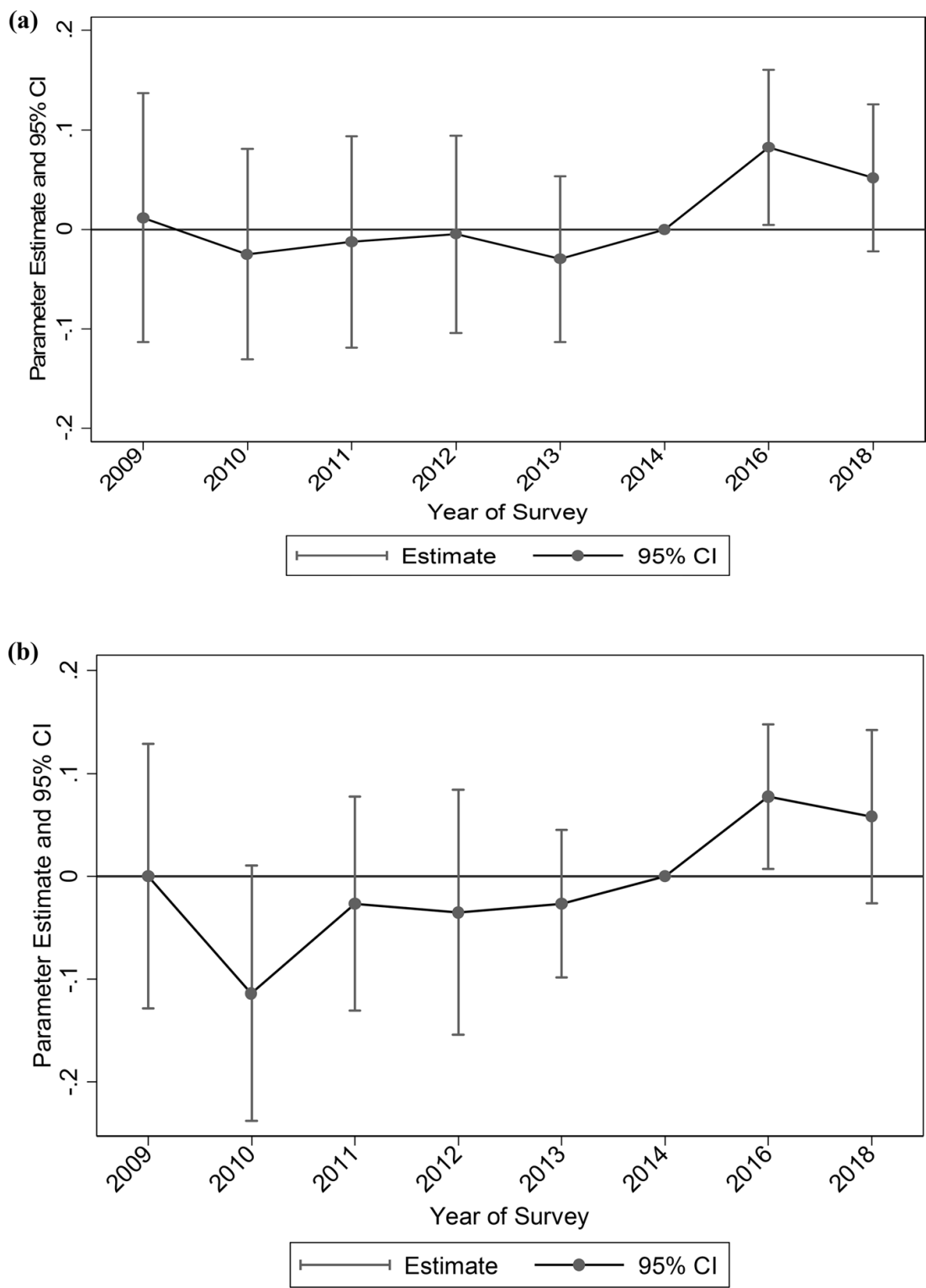


Fig. 3 Yearly effect of alcohol ban on reading and math test scores of children in Interior Bihar: Event study estimates. Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.041 and 0.171 for reading and 0.034 and 0.218 for math scores, respectively

test scores, respectively. The dots represent the estimated effects, and the bars represent the 95% confidence intervals. The figure shows that prior to the enactment of the policy in April 2016, the differences in the test scores (with respect to 2014) of children in Bihar and Jharkhand were very close to zero and statistically insignificant.

This lends support to our identifying assumption of parallel trends. After the alcohol prohibition act was implemented in Bihar in April 2016, the test scores of children in Bihar improved significantly compared to their peers in Jharkhand, where no such law was enacted. However, the effect was attenuated for the year 2018.²² This attenuation in effects is consistent with the documented trajectory of the ban's enforcement. As discussed in Section 2 and further noted by Chaudhuri et al. (2024) and Krishnatri and Vellakkal (2026), the prohibition was introduced in a highly stringent form in 2016, with strict enforcement and penalties, but was subsequently diluted in 2018 via amendments and relaxation of certain provisions. This timeline of implementation is reflected in two key facts: First, using data from the 68th and 79th rounds of the National Sample Survey (NSS) on consumption expenditure, we find an increase in the consumption of *toddy* (home-brewed liquor) over time (see Figure D.2 in Appendix). Second, an existing study by Chaudhuri et al. (2024) finds that violent crime in the neighborhood dips right after the ban and then rebounds. The timeline of our effects is consistent with both these key empirical facts regarding the timeline of implementation, alcohol consumption and violent crime in the neighborhood.

Recent advances in the econometric literature on difference-in-differences have raised concerns regarding the reliability of the event study designs in assessing the validity of the parallel trends assumption, owing to issues pertaining to statistical power. Therefore, following Rambachan and Roth (2023), we allow for the potential violations of the parallel trends and assess the sensitivity of our estimated effects to varying degrees to violations in the parallel trends. In other words, instead of assuming that the parallel trends assumption holds, we allow for progressively stronger departure from the parallel trends assumption and identify the threshold where our estimated effect becomes statistically indistinguishable from zero. We present the results in Appendix Figure A.1. Panels (a) and (b) present the results for two post periods for reading score and Panels (c) and (d) for math score. The results indicate that the effects for the first post treatment period (i.e., 2016) are robust to linear violations of parallel trends as well as progressively larger non-linear violations of parallel trends (see panels (a) and (c)). The breakdown points for the reading and math scores are 1.5% and 0.7% violation per year. The effects for 2018 are not robust to the violations of parallel trends (see panels (b) and (d)).

4.2 Treatment effect heterogeneity

In this section, we assess the heterogeneity in the treatment effects based on various characteristics of the child. More specifically, we investigate whether the alcohol ban has differential effects by the gender of the child, the socioeconomic background of the child and the education status of the father. These results provide crucial insights into potential mechanisms that may be driving the improvement in the learning outcomes post the alcohol ban.

We estimate a triple differences specification and present the results in Table 4. In Panel A, motivated by existing research by Petrusevich (2024), we check whether the alcohol ban has differential effects by the gender of the child. Columns 1 and 2,

²² The ASER test is typically conducted around November and December (Fagernäs & Pelkonen, 2020).

Table 4 Differential effects of alcohol ban on reading and math scores by child characteristics

	Standardized	
	Reading Score	Math Score
	(1)	(2)
<i>Panel A: Heterogeneity by Gender</i>		
Bihar Interior × Post	0.064* (0.034)	0.097*** (0.034)
Bihar Interior × Boy × Post	0.028 (0.024)	0.011 (0.022)
Observations	190,024	189,232
R-squared	0.471	0.478
<i>Panel B: Heterogeneity by Socioeconomic Status</i>		
Bihar Interior × Post	0.018 (0.030)	0.048 (0.032)
Bihar Interior × Rich × Post	0.053* (0.027)	0.061* (0.030)
Observations	190,024	189,232
R-squared	0.471	0.478
<i>Panel C: Heterogeneity by Father's Education</i>		
Bihar Interior × Post	0.037 (0.041)	0.044 (0.038)
Bihar Interior × Literate Father × Post	0.050 (0.038)	0.079** (0.031)
Observations	190,024	189,232
R-squared	0.472	0.479
District FE	Yes	Yes
Survey Year FE	Yes	Yes
Birth Cohort FE	Yes	Yes
Controls	Yes	Yes

Notes: The dependent variables are the reading and math scores, which have been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

respectively, show that while the policy has a positive differential effect on boys, the effects are not statistically different from zero. These results are in line with the findings of Petrusевич (2024) for health outcomes in Russia. Therefore, the mechanisms driving the results are plausibly gender-neutral.

Next, we check differential effects by the socioeconomic status of the child. Since ASER does not directly record the income of the household or its poverty status, we proxy for the income (poverty status) of children using the type of dwellings they reside in. Specifically, we assign children a high socioeconomic status if they reside in concrete or semi-concrete houses and those residing in houses made of mud or similar material are assigned a low socioeconomic status. We present results in Panel B of Table 4. Columns 1 and 2 show that the policy has no statistically significant effect on the sample of children belonging to poorer families in the interior districts of Bihar. However, the differential effects for relatively richer children are positive and statistically significant at the 10% level. These results suggest that the mechanisms driving these results could be a substitution of resources away from consumption from temptation goods (including alcohol) to education and nutrition of children.

Last, we investigate whether the policy has differential effects by the father's education. To this end, we interact a dummy, that takes the value 1 for a child whose father has been to school and 0 otherwise, with our coefficient of interest under a triple differences framework. We present the results in Panel C of Table 4. Interestingly, we find no statistically significant effects of the policy on the reading and math test scores of children whose fathers never went to school. However, the differential effect of the policy on children with literate fathers is positive and statistically significant for math score. This heterogeneity result points to the fact that the mechanisms plausibly driving these results could be an improvement in the household environment induced by the reduction of alcohol consumption on the part of adult males in the household.

4.3 Additional outcomes

Our primary focus in this paper is on children's learning outcomes. As mentioned in Section 1, this emphasis is motivated by the well-documented fact that India continues to face a severe learning crisis despite achieving near universal enrollment. Nevertheless, to provide a more comprehensive assessment of the effects of the alcohol prohibition policy of 2016, we present results from a set of additional education-related outcomes. Specifically, we estimate the impact of this policy change on (i) completed years of schooling, (ii) enrollment status, and (iii) dropout status. These outcomes capture the effect of this policy at the extensive margin and allow us to assess whether the policy affected participation alongside the improvements in learning outcomes.

For each of these outcomes, we estimate our baseline specification 1 and verify the validity of the parallel trends assumption using our event study specification 2. The results are presented in Appendix Table A.1. The dependent variables in columns 1, 2 and 3 are age-standardized years of schooling, (current) enrollment status and dropout status, respectively. We find no statistically significant effects on completed years of schooling and the probability of dropping out of school. However, we find a small decline in the probability of being currently enrolled after the ban. The magnitude of this decline is modest (about 1.4% of the sample mean).²³ The event study estimates, reported in Appendix Figures D.3, D.4 and D.5 confirm the absence of differential pre-trends of these outcomes, supporting the validity of the parallel trends assumption.

Taken together, these additional findings indicate that the effects of the alcohol prohibition on human capital formation does not primarily operate at the extensive margin via improvements in the schooling participation. Instead, the effects appear to manifest through improvements in learning conditional on enrollment. These results suggest that in a setting where baseline enrollment rates are already high and chances of dropping out are low, policy induced changes in household resource allocation and intra-household dynamics may influence the quality of educational engagement more than the decision of attend the school itself.

²³ Our main results are robust to using the sample of all children (including those currently enrolled and those that are not currently in school). Please refer to Table D.4 in the Appendix.

5 Potential mechanisms

Alcohol prohibition could affect children's test scores through at least three possible channels. These include improvement in the environment in the neighborhood, improvements in the home environment and parental behavior and increased expenditure on education and nutrition. We discuss these in detail below.

5.1 Improvements in the neighborhood environment

An extensive literature links alcohol to deterioration in the neighborhood quality via, for instance, an increase in violence and crime, suicides and road traffic collisions, among others (Hansen & Waddell, 2018; Lindo et al., 2018; Chaudhuri et al., 2024; Barron et al., 2024). Violence and crime, in turn, have been shown to have large adverse effects on children's cognitive outcomes both via direct exposure as well as via indirect exposure (Monteiro & Rocha, 2017; Michaelsen & Salardi, 2020; Shany, 2023). Moreover, living in an atmosphere of insecurity and fear induced by violence, children may suffer from huge mental health consequences. In a recent study, Ang (2020) shows that exposure to police violence in the United States increases the risk of emotional disturbance of children by 15% and reduces their GPA by 0.08 standard deviations.

In the context of the alcohol ban in Bihar, Chaudhuri et al. (2024) document that the policy led to a 0.22 standard deviation reduction in the violent crimes as compared to Jharkhand.²⁴ However, the reduction in violent crime was transient and it rebounded back one year after the policy was implemented. This temporal pattern closely mirrors our event study estimates, which show significant improvements in learning outcomes immediately after the ban but attenuation of the effect thereafter.

To provide further suggestive evidence, we use household consumption data from the 68th and 79th rounds of the NSSO surveys and document an increase over time in expenditure on *toddy* (home-brewed alcohol) in Bihar relative to Jharkhand (see Figure D.2) in the Appendix. Taken together, these pieces of evidence suggest that the initial gains in learning outcomes that we document may have been driven, at least in part, by the improvements in neighborhood safety and reduction in alcohol-related disruptions following the strict initial phase of the ban. As some of the most draconian provisions of the policy were watered down, enforcement weakened and informal alcohol consumption increased over time, these gains to improvements in the social environment may have dissipated, contributing to the attenuation of the learning gains afterwards.

5.2 Improvements in home environment and parental behavior

We start by investigating whether alcohol prohibition leads to an improvement in the home environment and parental behavior (especially the father's behavior). Alcohol-induced aggressive and violent behavior among parents (fathers, in our case)—such as intimate partner violence—may lead to deterioration in the quality of the home

²⁴ They do not find any effects for non-violent crime.

environment and make it less conducive for learning (Koenen et al., 2003; Bhuller et al., 2024). In fact, an extensive (medical) literature has empirically shown that alcohol consumption is strongly correlated with the incidence of domestic violence within households (Leonard & Quigley, 2017).

To test whether household environment quality improved after the prohibition of alcohol in Bihar, we investigate the effect of this policy on the consumption of alcohol by male partners/husbands, their aggressive behavior (such as incidence of intimate partner violence (IPV) and bargaining power of women within the household. Following Dixit and Mukherjee (2025) and Chakrabarti et al. (2024), we use data from the domestic violence module of the fourth and fifth rounds of the National Family Health Survey (NFHS-4 and -5)—the large-scale household-level surveys conducted in 2015-16 and 2019-21, respectively. We first construct indices for various types of violence against women using different questions—these capture emotional, physical and sexual violence and controlling behaviors by the husband. Next, we construct an aggregate index capturing overall intimate partner violence (IPV) by combining all three types of violence. We also investigate the effect on the bargaining power within households for ever-married women between the ages of 18 and 49. All the outcomes have been standardized with respect to the control group mean and standard deviation. Therefore, the coefficients can be interpreted as standard deviations. We discuss data sources and data construction in detail in Appendix B.

We estimate an equation similar to our baseline Eq. (1), in which our treatment group comprises ever-married women aged 18–49 from the interior districts of Bihar, and the control group comprises their counterparts in Jharkhand. Our Post dummy takes the value 1 for NFHS survey round 5 (2019-21) or 0 for NFHS round 4 (2015-16).²⁵ We present results in Table 5. Column 1 shows that the policy reduces the chance that a husband consumes alcohol. In terms of magnitude, post the prohibition, women in Bihar are about 0.08 standard deviations less likely to report that their husbands consume alcohol. The effect is statistically significant at the 5% level. Columns 2–4 show that women in Bihar are about 0.28–0.34 standard deviations less likely to experience emotional, physical and sexual violence after the enactment of the policy in April 2016 vis-à-vis their counterparts in Jharkhand. Column 5 shows a decline of 0.37 standard deviations in the overall incidence of IPV. The effect is statistically significant at the 1% level. In line with a reduction in overall incidence of IPV, Column 6 shows that women in Bihar are 0.46 standard deviations less likely to experience restrictions by their husbands and column 7 shows that the bargaining power of women within the households increased by about 0.26 standard deviations after the policy was enacted in Bihar.²⁶ In sum, we find significant improvements in the quality of the home environment and husband's aggressive and restrictive behavior towards their partners. These results are in line with the existing literature both in

²⁵ We restrict our sample to women in Bihar interviewed prior to Covid-19 and after the policy was enacted in April 2016.

²⁶ Husband exerts control over wife by restricting her contact with her female friend and family members, growing jealous when she interacts with other men, accusing her of being unfaithful and not trusting her with monetary matters.

Table 5 Effect of alcohol ban on the household environment and parental behavior

	Alcohol Consumption	Emotional Violence	Physical Violence	Sexual Violence	Overall IPV	Control- ling Behavior	Bar- gain- ing Power
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Bihar	-0.084**	-0.344***	-0.295***	-0.286***	-	-	0.256*
Interior					0.367***	0.460***	
× Post	(0.036)	(0.078)	(0.094)	(0.095)	(0.089)	(0.114)	(0.138)
District	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE							
NFHS	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave							
FE							
Con- trols	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obser- vations	6,351	6,351	6,351	6,351	6,351	6,308	5,994
R- squared	0.146	0.057	0.110	0.038	0.100	0.089	0.094

Notes: The dependent variables are alcohol consumption (a dummy taking value 1 if male partner consumes alcohol and 0 otherwise), indices for different types of violence faced by women (which has been standardized with respect to the control group mean and standard deviation so that the coefficients can be interpreted as standard deviations); the sample is restricted to women from Jharkhand and the interior districts of Bihar; individual and household level controls include respondent’s age, respondent’s education, respondent’s employment status, husband’s education, number of years of marriage, religion, caste, household size, age of household head, gender of household head, wealth index, number of children less than 5 years in the household, respondent’s justification of wife beating on being unfaithful and respondent’s autonomy; data source is the NFHS-4 (2015-16) and NFHS-5 (2019-2022); standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

the context of India and other countries like Mexico (Angelucci, 2008; Khurana & Mahajan, 2022; Chakrabarti et al., 2024; Dixit & Mukherjee, 2025).

Since identification in a difference-in-differences framework heavily relies on the parallel trends assumption, it is important to assess whether the pre-policy trends in these outcomes evolved similarly across Bihar and Jharkhand. Unfortunately, the two most recent waves of NFHS do not provide enough data points to implement an event study analysis. However, to show supportive evidence for the common trends assumption, we follow Dixit and Mukherjee (2025) and turn to administrative data from the National Crime Records Bureau (NCRB). Specifically, we examine trends in reported incidences of cruelty against women by husbands or relatives (a widely proxy for domestic violence). We present the results in Appendix Figure A.2 which shows that these reports of cruelty against women by husbands or relatives were evolving in parallel in Bihar and Jharkhand prior to the implementation of ban. Following the ban, reported cases declined sharply in Bihar vis-à-vis Jharkhand. The absence of this differential pre-trends in the closely related proxy for domestic violence provides support for the validity of parallel trends assumption underlying our main estimates in Table 5.

5.3 Expenditure on education and nutrition

Another potential mechanism that could link the alcohol ban to improvements in learning outcomes is an increase in educational expenditures due to the substitution effect. After the ban on alcohol, households may shift their expenditure from temptation goods (including alcohol) to more productive uses such as children's education and nutrition. Since lower domestic violence could lead to greater maternal control and bargaining power within the household, these households may spend more on the educational and nutrition needs of their children.²⁷ Recent papers (Falézan, 2020; Mukherji & Chakraborty, 2024; Krishnatri & Vellakkal, 2026) show that expenditures on alcohol significantly decreased with respect to the control state of Jharkhand. Falézan (2020) shows that after the 2016 ban on alcohol, households in Bihar reduced expenditures on alcohol and other temptation goods and increased expenditures on education and nutritious food. The paper finds a larger reduction in alcohol expenditures among richer households. Further, Mukherji and Chakraborty (2024) present evidence that the ban led to an increase in educational expenditure and to a reallocation of healthcare spending from urgent medical expenses towards positive lifestyle changes. Krishnatri and Vellakkal (2026) also find that the alcohol ban in Bihar increased caloric, protein, and fat intake from healthy food sources and reduced fat intake from unhealthy sources by reallocating the income previously spent on alcohol.

While the ASER surveys do not collect data on educational expenditure to directly test this mechanism, it does contain information on whether the child takes after-class paid private tuition. In the Indian context, paid tuition/coaching classes for a small group of students led by a private tutor are popular. These coaching sessions typically take place outside the school and after class hours. Since availing paid private tuition is a choice, whether the child avails paid private tuition can be thought of as a proxy for educational spending and parents' seriousness regarding the education of their children. Using the ASER data, we investigate how the alcohol ban affected the probability of availing paid private tuition. Table 6 presents the result from estimating specification (1).²⁸ Column 1 shows that the policy increased the probability of availing paid private tuition.²⁹ Columns 2–5 show that the positive effect is robust to the inclusion of district fixed effect (Column 2), survey year fixed effects (Column 3), birth cohort fixed effects (Column 4) and a bunch of individual and household characteristics (Column 5). Overall, using our preferred specification, we find an increase of 4.3% points in the probability of children in the interior districts of Bihar availing paid private tuition after the policy, compared to their counterparts in the control group. In line with the findings of Falézan (2020) and Mukherji and Chakraborty

²⁷ An extensive literature has shown that women tend to spend more of their children's education and health (Lundberg et al., 1997; Smith et al., 2003; Bobonis, 2009; Fafchamps et al., 2009; Baranov et al., 2020).

²⁸ We present the event study estimates in the Appendix Figure A.3.

²⁹ The term "tuition" here refers to after-school private coaching classes outside of regular school hours. Such after-class coaching sessions are common in India. The term "tuition" does not refer to school tuition fees.

Table 6 Effect of alcohol ban on uptake of after-class paid private tuition

	After-class Private Tuition				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior × Post	0.042** (0.017)	0.041** (0.016)	0.042** (0.016)	0.039** (0.016)	0.043** (0.017)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	232,466	232,466	232,466	232,217	174,002
R-squared	0.076	0.122	0.123	0.134	0.180

Notes: The dependent variable is after-class paid private tuition, a dummy that takes value 1 if the child goes to private tuition classes and 0 otherwise; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child’s age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * p < 0.1, ** p < 0.05, *** p < 0.01

(2024), these results indicate that educational expenditure increased post the alcohol ban in Bihar.

5.4 Additional potential mechanisms

An extensive strand of literature in education highlights the importance of teacher presence for student learning. For instance, Duflo et al. (2012) show experimentally that reducing teacher absenteeism through monitoring and incentive mechanisms leads to significant improvements in student test scores. Das et al. (2007) document that high teacher absenteeism in Zambia significantly reduces test score gains. Taken together, this literature provides strong evidence that instructional time—proxied by teacher attendance—is a critical determinant of learning outcomes.³⁰

Motivated by this evidence, we examine whether changes in school attendance (for both the teacher and student) could explain the improvement in learning that we document as a result of alcohol prohibition policy. To this end, we use ASER school-level data and estimate our baseline specification (1) using student and teacher attendance as outcome variables. We present these results in Appendix Tables A.2 and A.3. The results indicate that both the teacher and student attendance decline in Bihar relative to Jharkhand in the post-ban period. However, the event study estimates for these outcome variables, presented in Appendix Figures D.6 and D.7 reveal significant differential pre-trends. Given this violation of the parallel trends assumption, we interpret these findings with caution and do not assign them a causal interpretation.

Nonetheless, based on the literature cited above, lower attendance on part of both the teachers and students should lead to a deterioration in children’s learning outcomes. The fact that we observe improvements in learning despite declines in attendance suggests that the gains documented are unlikely to be driven by improvements in school-level inputs. Our results indicate that the benefits of a better social environ-

³⁰ Reduced instructional time may also impact test scores via student absenteeism.

ment and greater spending on education over-ride any potential losses from lower teacher or student attendance.

Another consequence of the alcohol ban was that it would adversely affect families whose livelihoods depended on the alcohol industry. The alcohol ban led to the closure of 21 alcohol manufacturing plants, and is estimated to have cost 35,000 direct jobs (Muralidharan, 2024, (Ch. 7, pg. 223). These estimated job losses are miniscule relative to the size of the workforce. For context, note that Bihar's (projected) population in 2023 was more than 120 million (comparable to the population of Japan, Mexico and Ethiopia) (Muralidharan, 2024, Table 8.1 in Ch. 8, pages 237–38). Nonetheless, it bears emphasis that the estimated learning gains observed in the data are observed despite these job losses. Therefore, the gains that we report are net of the losses.

6 Robustness checks

In this section, we present a series of robustness checks to further validate our main results.

6.1 Potential confounders

We start by conducting a few checks to ensure that our identification strategy captures the effects of the alcohol ban rather than the effects due to confounding factors. To this end, we first divide our sample into districts with high baseline alcohol consumption and districts with low alcohol consumption. Based on the NFHS-4 data, we create a dummy that takes value 1 for children in districts with high baseline alcohol consumption (above 25th percentile or above average consumption in the state) and 0 otherwise. If our results are driven by an alcohol ban, then we should see very modest or muted effects for children in the districts with low baseline alcohol consumption. We present results in Table 7. Columns 1 and 3 show that the improvements in learning outcomes are actually concentrated among children from districts with high baseline alcohol consumption, and the effects for children in the low alcohol consumption districts are small in magnitude and statistically insignificant.³¹

To further support our identification strategy, we check the heterogeneity in the treatment effects by religion. Since alcohol consumption is considered a major sin according to Islamic jurisprudence and is forbidden, a majority of Indian Muslims do not drink alcohol (Bennett et al., 1998).³² Therefore, if these are the effects of alcohol ban, we would expect muted effects on the sample of children from Muslim families. Unfortunately, ASER does not record the religion of the child, and we cannot check heterogeneity by religion. However, following the classification of districts in Chaudhuri et al. (2024), we divide our sample along the lines of religion based on

³¹ Appendix Table A.4 further shows that these results hold if we define high and low baseline alcohol status based on average alcohol consumption in the state.

³² In the state of Bihar, while 38% of Hindu women report that their husbands drink alcohol, only about 18% of Muslim women report the same (NFHS-4).

Table 7 Heterogenous effect of alcohol ban on reading and math scores by baseline alcohol consumption

	Standardized			
	Reading Score		Math Score	
	Consumption		Consumption	
	High	Low	High	Low
	(1)	(2)	(3)	(4)
Bihar Interior × Post	0.092*** (0.027)	-0.018 (0.027)	0.116*** (0.028)	0.014 (0.052)
District FE	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	178,262	112,540	177,560	112,077
R-squared	0.472	0.480	0.477	0.484

Notes: The dependent variables are reading and math scores, which have been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; high consumption districts constitute all districts with baseline alcohol consumption above 25th percentile and the opposite is true for low consumption districts; individual and household level controls include child’s age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

the baseline Muslim population in a district. Districts with a higher proportion of Muslims with respect to the population of Muslims in Bihar are considered Muslim-majority districts. We present the results in Table 8. As expected, we find that the effects are driven only by children in Hindu majority districts. The effects for children in Muslim-majority regions are smaller in magnitude and statistically insignificant.

6.2 Alternative clustering and the exact randomization test

Following a series of existing studies (see Chaudhuri et al. (2024); Dixit and Mukherjee (2025) and Krishnatri and Vellakkal (2026) on the Bihar alcohol ban, we cluster standard errors at the district level in our main analysis. To check the robustness of our results to alternative ways of clustering, we re-run our baseline regression (1) and cluster standard errors in several different ways. For instance, we cluster at the levels of state, state-year, state-cohort, district, district-year and age cohorts. We use wild cluster bootstrapped standard errors whenever the number of clusters are less than 50. The results are presented in the Appendix Tables D.1, D.2 and D.3. Across specifications, we find result that are consistent with our main findings in Tables 2, 3 and 6.

As a further robustness check, we complement our cluster-based and wild-cluster bootstrapped based inference with exact randomization tests (Bharadwaj et al., 2014). This approach to inference directly relies on the treatment assignment mechanism and does not rely on cluster asymptotics (Young, 2019). We randomly assign treatment status in the data and estimate our baseline Eq. (1). We repeat this exercise 1,000 times and collect these 1,000 placebo estimates of the treatment effect. We then plot the distribution of the 1,000 placebo estimates and compare them with our observed estimates. The results presented in Appendix Figure D.8 show that, relative

Table 8 Heterogenous effect of alcohol ban on reading and math scores by religious majority

	Standardized			
	Reading Score		Math Score	
	Muslim Majority	Hindu Majority	Muslim Majority	Hindu Majority
	(1)	(2)	(3)	(4)
Bihar Interior × Post	0.051 (0.035)	0.091*** (0.029)	0.059 (0.043)	0.127*** (0.027)
District FE	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	132,505	158,297	131,974	157,663
R-squared	0.476	0.473	0.482	0.477

Notes: The dependent variables are reading and math scores, which have been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; districts with a higher proportion of Muslims with respect to the population of Muslims in Bihar are considered Muslim-majority districts as classified in (Chaudhuri et al., 2024); individual and household level controls include child’s age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

to the distribution of placebo effect sizes, our actual estimates are extreme outliers. This indicates that it is high unlikely to observe the effects that we find just by chance.

6.3 Synthetic control method

In Section 3, we discussed why Jharkhand is an appropriate control group for Bihar. Moreover, we validated the assumption of parallel trends and showed the robustness of our estimates in the case of a potential violation of parallel trends in Section 4. However, one may still argue that the choice of the control group is somewhat arbitrary. Therefore, as a robustness check, we take a bunch of neighboring states as our potential control group and use the Synthetic Control Method to construct the optimal control group. These control states are Jharkhand, West Bengal, Uttar Pradesh, Odisha, Chhattisgarh and Madhya Pradesh. Appendix Table A.5 shows the synthetic control weights for each of the states in the construction of the counterfactual for Bihar.³³ We present results from the synthetic control method in Appendix Figure A.4. We find significant improvement in both the reading and math scores after the alcohol prohibition act was enacted in 2016. Prior to 2016, the actual and synthetic outcome paths almost overlapped. However, starting in 2016, we see a divergence in the outcome paths, and the actual path is above the counterfactual path.

³³ Appendix Table A.6 shows the covariate balance between the actual and synthetic versions of Bihar.

6.4 Alternative samples and specifications

To further test the robustness of our results, we extend our sample to include border districts of Bihar, children not currently in school and all children irrespective of enrollment status. First, we expand our treatment group to also include the border districts of Bihar. We present the results in Appendix Figure A.5. We find very similar results. The event study graphs show that the reading and math scores improved in 2016, and for 2018, the effects are not statistically different from zero. Moreover, we find that prior to 2016, the test scores of children in all of Bihar and Jharkhand were trending similarly. Second, we expand our sample to include not currently enrolled (drop-out) and never enrolled children. We present the results in Appendix Table D.4 and Figures D.9 and D.10. The results are consistent with our main findings in Tables 2 and 3. Next, we show that our estimates are robust to a range of alternate specifications. We start by estimating our main specification using an alternative outcome measure in which the test scores are standardized within birth cohorts (i.e., age-standardized),³⁴ rather than standardizing over the full sample. This accounts for concerns related to potential differences in age distribution across survey waves and treatment and control groups in our repeated cross-sections. The results, presented in Appendix Tables D.5 and D.6, are in line with our baseline results reported in Tables 2 and 3.³⁵

Since we lack controls at the geographic level such as local economic conditions or institutional changes that may vary over time, in the absence of state-by-year fixed effects, these can be absorbed into the error term in our baseline specification. To address this concern, we estimate an additional specification including state-specific linear time trends. This approach allows each state to trend differently over time. The results, reported in Appendix Tables D.7 and D.8, are broadly consistent with our main findings. In the richest specification, the coefficient of interest remains positive and statistically significant at the 10% level for reading, while the estimate for math is positive but not statistically significant at conventional levels. Given the limited number of post-treatment periods, disentangling treatment effects from state-specific trends is inherently challenging; nevertheless, the overall pattern of results remains stable and the coefficients for first post-treatment period 2016 remains statistically significant at the 95% level (See Appendix Figure D.12).

Lastly, we present results from the linear probability models for each of the eight components of reading and math tests in Appendix Table A.7. The dependent variables are dummies, which take value 1 if the child has achieved the level of mastery as indicated by the model label and zero otherwise. Panels A and B show that alcohol prohibition increased the likelihood that a child is able to read at different levels of mastery, recognize numbers, and perform subtraction and division problems. The effects are statistically significant. The results from this alternative specification align with our main results and alleviate concerns regarding measurement errors in our outcome variables.

³⁴ We standardize the reading and math test scores within each cohort with respect to the mean and standard deviation of the control group.

³⁵ Appendix Figure D.11 reports the associated event study results.

6.5 Propensity score matching

To address potential concerns related to differences in observables characteristics between the children in the treated districts of Bihar vis-à-vis their counterparts in Jharkhand, we implement a propensity score matching (PSM) procedure combined with our main difference-in-differences estimation (PSM-DID). We start by estimating the propensity scores from a probit model predicting the probability of residing in the treated districts of Bihar as a function of child and household-level characteristics observed over multiple time periods. Then, we implement a matching difference-in-difference model, where we match based on the estimated propensity scores using an Epanechnikov kernel.

We present the results from the PSM-DID estimation in Appendix D. Table D.9 shows the balance of covariates before and after matching. Figure D.13 plots the kernel density of propensity scores before and after matching. Tables D.10 and D.11 show the results from the weighted difference-in-difference regressions. The results are consistent with our baseline estimates in Tables 2 and 3.

7 Cost-benefit analysis

While a full cost-benefit analysis of the Bihar Alcohol Prohibition Act (2016) is beyond the scope of this paper, we provide a rough back-of-the-envelope calculation of the costs and benefits. Our cost-benefit analysis accounts only for fiscal costs of the ban and only those benefits due to improved learning outcomes. We estimate the translation of improved learning outcomes among exposed children into expected benefits in terms of their future income. Our results reflect an improvement of 0.074 and 0.102 standard deviations in the reading and arithmetic abilities of children in post-ban Bihar, implying an average improvement of 0.088 standard deviations in their learning abilities.

Using two consecutive rounds of the Indian Human Development Survey (Desai et al., 2008; Desai & Vanneman, 2015) conducted in 2004-05 and 2011-12, respectively, we find that, on average, a one standard deviation improvement in learning abilities results in an additional 1.5 completed years of schooling.³⁶ Following this, our representative children receive 0.132 years of additional schooling in the post-ban Bihar. Fulford (2014) finds 5.15% returns to an additional year of schooling.³⁷ In line with this, an additional 0.132 years of schooling yield returns of about 0.68% to children exposed to the ban.

In 2016, Bihar's population in the age group of 16–17 was 3,471,483 (Government of India, 2016). The Net State Domestic Product per capita at 2011-12 constant prices was 24,064 rupees in 2016 (CEIC, 2024). With a compounded annual growth

³⁶ We estimated that a one standard deviation increase in reading and math score of a child aged 11 years in IHDS-1 is associated with an additional 1.5 and 1.4 year of schooling respectively, at the age of 17 in IHDS-2.

³⁷ Following Fulford (2014), men have 4.6% returns to an additional year of education in the casual wage market and 5.8% in the salary market. For our calculation, here we take an average of both of these returns.

rate of 2% for Net State Domestic Product and the discount rate of 5.75%,^{38,39} assuming that this group of children will work for the next 30 years, then the total present discounted value of the additional income these exposed children earn due to this policy change is 9,636 million rupees.⁴⁰ This estimated benefit from the spillover effects of Bihar's alcohol ban is almost one-third of the lost revenue of 31,417 million rupees in the year 2016. We should re-emphasize that our cost-benefit analysis ignores benefits of the policy that materialize through channels besides improved learning outcomes. It bears emphasis that recent studies have identified multiple benefits of the policy including lower crimes, reduced intimate partner violence, healthier lifestyle, reduced tobacco consumption, and greater spending on healthier food items (Chaudhuri et al., 2024; Dixit & Mukherjee, 2025; Chakrabarti et al., 2024; Falézan, 2020; Krishnatri & Vellakkal, 2026). Therefore, our analysis likely understates the benefits of the policy. Still, we find even the benefits from just improved learning outcomes to be a substantial fraction of costs.

8 Conclusion

We investigate the effect of alcohol prohibition on children's learning outcomes. The context is the Indian state of Bihar, which implemented a complete ban on the manufacturing, sale, transport and consumption of alcohol in April 2016. We exploit a plausibly exogenous variation in the availability of alcohol generated by this policy using a difference-in-differences framework. Using data on reading and math test scores from 2009 to 2018, we find that the ban improved the reading and math scores by 0.076 and 0.102 standard deviations, respectively. Further, we show that the ban improved the quality of the home environment, such as by reducing alcohol consumption and intimate partner violence and by improving the relative position of women in the household. Moreover, we find evidence of increased expenditures on education as a result of the policy. We conduct a battery of robustness checks to validate our results. Our results indicate that eliminating the negative externalities associated with alcohol consumption within the household—such as the incidence of domestic violence, parental ill behavior and substitution of household resources away from productive uses to expenditures on temptations goods—can result in significant gains in children's cognition. From a policy perspective, these results indicate that simple and cost-efficient policies such as alcohol bans can prove to be effective in improving the learning outcomes of children in India.

While this study documents significant gains in learning among children in Bihar post the alcohol prohibition policy, we acknowledge few limitations of the study that warrant consideration. First, the ASER test scores used in the analysis measure basic foundational literacy and numeracy skills and are not grade-appropriate. While we restrict the analysis to primary and middle school children for whom these tests are somewhat more relevant, we are limited in scope to fully capture the changes in

³⁸ Authors' calculations from CEIC Data.

³⁹ The reverse repo rate for government borrowings in 2016-17 (Reserve Bank of India, 2016).

⁴⁰ For detailed calculations, please see Appendix C.

curriculum-based achievement. Second, our cost-benefit analysis is a rough back-of-the-envelope calculation intended to contextualize the spillover effects of the prohibition policy. A comprehensive welfare analysis of the policy is an open problem that is on our agenda for future research.

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Declarations

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References

- Andreas, J. B., Torvik, F. A., & Lund, I. O. (2022). Parental binge drinking and offspring's high school non-completion: a prospective HUNT survey and educational registry study. *Drug and Alcohol Dependence*, 230, 109189. <https://doi.org/10.1016/j.drugalcdep.2021.109189>
- Ang, D. (2020). The effects of police violence on inner-city students*. *The Quarterly Journal of Economics*, 136(1), 115–168. <https://doi.org/10.1093/qje/qjaa027>
- Angelucci, M. (2008). Love on the rocks: domestic violence and alcohol abuse in rural Mexico. *The BE Journal of Economic Analysis & Policy*, 8(1).
- ASER. (2019). *Annual Status of Education Report (Rural), 2009–2018*. Pratham.
- Banerjee, A. V., & Duflo, E. (2005). Growth theory through the lens of development economics. *Handbook of Economic Growth*, 1, 473–552. [https://doi.org/10.1016/S1574-0684\(05\)01007-5](https://doi.org/10.1016/S1574-0684(05)01007-5). Elsevier.
- Baranov, V., Bhalotra, S., Biroli, P., & Maselko, J. (2020). Maternal depression, women's empowerment, and parental investment: evidence from a randomized controlled trial. *American Economic Review*, 110(3), 824–859. <https://doi.org/10.1257/aer.20180511>
- Barham, T., Macours, K., & Maluccio, J. A. (2013). More schooling and more learning? effects of a three-year conditional cash transfer program in Nicaragua after 10 years, Working paper.
- Barron, K., Parry, C. D. H., Bradshaw, D., Dorrington, R., Groenewald, P., Laubscher, R., & Matzopoulos, R. (2024). Alcohol, violence, and injury-induced mortality: evidence from a modern-day prohibition. *The Review of Economics and Statistics*, 106(4), 938–955. https://doi.org/10.1162/rest_a_01228
- Baughman, R., Conlin, M., Dickert-Conlin, S., & Pepper, J. (2001). Slippery when wet: the effects of local alcohol access laws on highway safety. *Journal of Health Economics*, 20(6), 1089–1096. [https://doi.org/10.1016/S0167-6296\(01\)00103-5](https://doi.org/10.1016/S0167-6296(01)00103-5)
- Benhassine, N., Devoto, F., Duflo, E., Dupas, P., & Pouliquen, V. (2015). Turning a shove into a nudge? a labeled cash transfer for education. *American Economic Journal: Economic Policy*, 7(3), 86–125. <https://doi.org/10.1257/pol.20130225>
- Bennett, L. A., Campillo, C., Chandrashekar, C. R., & Gureje, O. (1998). Alcoholic beverage consumption in India, Mexico, and Nigeria: a cross-cultural comparison. *Alcohol Health and Research World*, 22(4), 243–252.
- Bharadwaj, P., Johnsen, J. V., & Løken, K. V. (2014). Smoking bans, maternal smoking and birth outcomes. *Journal of Public Economics*, 115, 72–93. <https://doi.org/10.1016/j.jpubeco.2014.04.008>

- Bhuller, M., Dahl, G. B., Løken, K. V., & Mogstad, M. (2024). Domestic violence reports and the mental health and well-being of victims and their children. *Journal of Human Resources*, 59(S), S152–S186. <https://doi.org/10.3368/jhr.1222-12698R1>
- Biderman, C., De Mello, J. M. P., & Schneider, A. (2010). Dry laws and homicides: evidence from the São Paulo metropolitan area. *The Economic Journal*, 120(543), 157–182. <https://doi.org/10.1111/j.1468-0297.2009.02299.x>
- Bobonis, G. J. (2009). Is the allocation of resources within the household efficient? new evidence from a randomized experiment. *Journal of Political Economy*, 117(3), 453–503. <https://doi.org/10.1086/600076>
- Borkum, E., He, F., & Linden, L. L. (2012). School libraries and language skills in Indian primary schools: a randomized evaluation of the Akshara library program. Working Paper No. 18183. *National Bureau of Economic Research*.
- Caces, M. F., Harford, T. C., Williams, G. D., & Hanna, E. Z. (1999). Alcohol consumption and divorce rates in the United States. *Journal of Studies on Alcohol*, 60(5), 647–652. <https://doi.org/10.15288/jsa.1999.60.647>
- Carpenter, C. (2004). Heavy alcohol use and youth suicide: evidence from tougher drunk driving laws. *Journal of Policy Analysis and Management*, 23(4), 831–842. <https://doi.org/10.1002/pam.20049>
- Carpenter, C. (2005a). Youth alcohol use and risky sexual behavior: evidence from underage drunk driving laws. *Journal of Health Economics*, 24(3), 613–628. <https://doi.org/10.1016/j.jhealeco.2004.09.014>
- Carpenter, C. S. (2005b). Heavy alcohol use and the commission of nuisance crime: evidence from underage drunk driving laws. *American Economic Review*, 95(2), 267–272.
- Carpenter, C. (2007). Heavy alcohol use and crime: evidence from underage drunk-driving laws. *The Journal of Law and Economics*, 50(3), 539–557. <https://doi.org/10.1086/519809>
- Carpenter, C., & Dobkin, C. (2009). The effect of alcohol consumption on mortality: regression discontinuity evidence from the minimum drinking age. *American Economic Journal: Applied Economics*, 1(1), 164–182. <https://doi.org/10.1257/app.1.1.164>
- Carrell, S. E., & Hoekstra, M. L. (2010). Externalities in the classroom: how children exposed to domestic violence affect everyone's kids. *American Economic Journal: Applied Economics*, 2(1), 211–228. <https://doi.org/10.1257/app.2.1.211>
- CEIC (2024). NSDP Per Capita: 2011–12p: Bihar. <https://www.ceicdata.com/en/india/net-state-domestic-product-per-capita/nsdp-per-capita-201112p-bihar>
- Chakrabarti, S., Christopher, A., Scott, S., Kishore, A., & Nguyen, P. H. (2024). Effects of a large-scale alcohol ban on population-level alcohol intake, weight, blood pressure, blood glucose, and domestic violence in India: a quasi-experimental population-based study. *The Lancet Regional Health - South-east Asia*, 26, 100427. <https://doi.org/10.1016/j.lansea.2024.100427>
- Chakraborty, T., & Jayaraman, R. (2019). School feeding and learning achievement: evidence from India's midday meal program. *Journal of Development Economics*, 139, 249–265. <https://doi.org/10.1016/j.jdeveco.2018.10.011>
- Chaudhuri, K., Jha, N., Nilayamgode, M., & Suryanarayana, R. (2024). Alcohol ban and crime: the abcs of the Bihar prohibition. *Economic Development and Cultural Change*, 72(4), 1795–1827. <https://doi.org/10.1086/725452>
- Collins, R. L., Ellickson, P. L., & Klein, D. J. (2007). The role of substance use in young adult divorce. *Addiction*, 102(5), 786–794. <https://doi.org/10.1111/j.1360-0443.2007.01803.x>
- Dahl, G. B., & Lochner, L. (2012). The impact of family income on child achievement: evidence from the earned income tax credit. *American Economic Review*, 102(5), 1927–1956. <https://doi.org/10.1257/aer.102.5.1927>
- Dar, A., & Sahay, A. (2018). Designing policy in weak states: unintended consequences of alcohol prohibition in Bihar. *Available at SSRN*, 3165159. <https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=3165159>
- Das, J., Dercon, S., Habyarimana, J., & Krishnan, P. (2007). Teacher shocks and student learning: evidence from Zambia. *Journal of Human Resources*, 42(4), 820–862. <https://doi.org/10.3368/jhr.42.4.820>
- De Ree, J., Muralidharan, K., Pradhan, M., & Rogers, H. (2018). Double for nothing? experimental evidence on an unconditional teacher salary increase in Indonesia*. *The Quarterly Journal of Economics*, 133(2), 993–1039. <https://doi.org/10.1093/qje/qjx040>
- Deming, D. J. (2022). Four facts about human capital. *Journal of Economic Perspectives*, 36(3), 75–102. <https://doi.org/10.1257/jep.36.3.75>
- Desai, S., & Vanneman, R. (2015). *India Human Development Survey-II (IHDS-II), 2011–12*: ICPSR - Interuniversity Consortium for Political and Social Research. <https://doi.org/10.3886/ICPSR36151.V6>

- Desai, S., Vanneman, R., & National Council Of Applied Economic Research, New Delhi. (2008). *India Human Development Survey (IHDS), 2005: Version 12*. ICPSR - Interuniversity. <https://doi.org/10.3886/ICPSR22626.V12>. Consortium for Political and Social Research.
- Dixit, M., & Mukherjee, S. (2025). Alcohol prohibition and intimate partner violence: evidence from an Indian state. *Economic Development and Cultural Change*, 74(1), 37–84. <https://doi.org/10.1086/734542>
- Duflo, E. (2001). Schooling and labor market consequences of school construction in Indonesia: evidence from an unusual policy experiment. *American Economic Review*, 91(4), 795–813. <https://doi.org/10.1257/aer.91.4.795>
- Duflo, E., Hanna, R., & Ryan, S. P. (2012). Incentives work: getting teachers to come to school. *American Economic Review*, 102(4), 1241–1278. <https://doi.org/10.1257/aer.102.4.1241>
- Evans, D. K., & Yuan, F. (2022). How big are effect sizes in international education studies? *Educational Evaluation and Policy Analysis*, 44(3), 532–540. <https://doi.org/10.3102/01623737221079646>
- Fafchamps, M., Kebede, B., & Quisumbing, A. R. (2009). Intrahousehold welfare in rural Ethiopia*. *Oxford Bulletin of Economics and Statistics*, 71(4), 567–599. <https://doi.org/10.1111/j.1468-0084.2009.00553.x>
- Fagnäs, S., & Pelkonen, P. (2020). Teachers, electoral cycles, and learning in India. *Journal of Human Resources*, 55(2), 699–732. <https://doi.org/10.3368/jhr.55.3.0816-8100R>
- Falézan, C. (2020). Expenditures, savings and temptation under prohibition evidence from Bihar. *Unpublished Manuscript*. <https://nrs.harvard.edu/URN-3:HUL.INSTREPOS:37364699>
- Fulford, S. (2014). Returns to education in India. *World Development*, 59, 434–450. <https://doi.org/10.1016/j.worlddev.2014.02.005>
- Gershenson, S., & Tekin, E. (2018). The effect of community traumatic events on student achievement: evidence from the Beltway sniper attacks. *Education Finance and Policy*, 13(4), 513–544. https://doi.org/10.1162/edfp_a_00234
- Gil-González, D., Vives-Cases, C., Álvarez-Dardet, C., & Latour-Pérez, J. (2006). Alcohol and intimate partner violence: do we have enough information to act? *European Journal of Public Health*, 16(3), 278–284. <https://doi.org/10.1093/eurpub/ckl016>
- Glewwe, P., & Muralidharan, K. (2022). Improving School Education Outcomes in Developing Countries. *RISE WP No. 15/001*. University of Oxford. <https://doi.org/10.25446/oxford.21107362.v1>
- Government of India (2016). Population projection in 6–23 years of age group. Technical report, Ministry of Education. https://www.education.gov.in/statistics-new?shs_term_node_tid_depth=389&Apply=Apply
- Green, C., & Krehic, L. (2022). An extra hour wasted? bar closing hours and traffic accidents in Norway. *Health Economics*, 31(8), 1752–1769. <https://doi.org/10.1002/hec.4550>
- Green, C. P., Heywood, J. S., & Navarro, M. (2014). Did liberalising bar hours decrease traffic accidents? *Journal of Health Economics*, 35, 189–198. <https://doi.org/10.1016/j.jhealeco.2014.03.007>
- Hansen, B., & Waddell, G. R. (2018). Legal access to alcohol and criminality. *Journal of Health Economics*, 57, 277–289. <https://doi.org/10.1016/j.jhealeco.2017.08.001>
- Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of Economic Literature*, 46(3), 607–668. <https://doi.org/10.1257/jel.46.3.607>
- Heaton, P. (2012). Sunday liquor laws and crime. *Journal of Public Economics*, 96(1), 42–52. <https://doi.org/10.1016/j.jpubeco.2011.08.002>
- Hidalgo, D., Onofa, M., Oosterbeek, H., & Ponce, J. (2013). Can provision of free school uniforms harm attendance? evidence from Ecuador. *Journal of Development Economics*, 103, 43–51. <https://doi.org/10.1016/j.jdeveco.2012.12.006>
- Houtenville, A. J., & Conway, K. S. (2008). Parental effort, school resources, and student achievement. *Journal of Human Resources*, 43(2), 437–453. <https://doi.org/10.3368/jhr.43.2.437>
- <https://doi.org/10.1257/000282805774670220>
- <https://iris.who.int/bitstream/handle/10665/274603/9789241565639-eng.pdf;2018>
- <https://repositorio.minedu.gob.pe/handle/20.500.12799/3371>
- Jolex, A., & Kaluwa, B. (2022). Crowding out effects of alcohol consumption expenditure on household resource allocation in Malawi. *PLOS ONE*, 17(2), e0263330. <https://doi.org/10.1371/journal.pone.0263330>
- Khurana, S., & Mahajan, K. (2022). Public safety for women: is regulation of social drinking spaces effective? *The Journal of Development Studies*, 58(1), 164–182. <https://doi.org/10.1080/00220388.2021.1961747>

- Koenen, K. C., Moffitt, T. E., Caspi, A., Taylor, A., & Purcell, S. (2003). Domestic violence is associated with environmental suppression of IQ in young children. *Development and Psychopathology*, 15(2), 297–311. <https://doi.org/10.1017/S0954579403000166>
- Krämer, M., Gautam, K., S., & Vollmer, S. (2021). Improving child health and cognition: evidence from a school-based nutrition intervention in India. *Review of Economics and Statistics*, 103(5), 818–834.
- Krishnatri, V., & Vellakkal, S. (2026). Does alcohol prohibition improve caloric and macronutrient intake from healthy food sources? evidence from Bihar, India. *Agricultural Economics*, 57(1), e70090. <https://doi.org/10.1111/agec.70090>
- Lahoti, R., & Sahoo, S. (2020). Are educated leaders good for education? evidence from India. *Journal of Economic Behavior & Organization*, 176, 42–62. <https://doi.org/10.1016/j.jebo.2020.03.026>
- Leonard, K. E., & Quigley, B. M. (2017). Thirty years of research show alcohol to be a cause of intimate partner violence: future research needs to identify who to treat and how to treat them. *Drug and Alcohol Review*, 36(1), 7–9. <https://doi.org/10.1111/dar.12434>
- Lindo, J. M., Siminski, P., & Swensen, I. D. (2018). College party culture and sexual assault. *American Economic Journal: Applied Economics*, 10(1), 236–265. <https://doi.org/10.1257/app.20160031>
- Lowthian, E. (2022). The secondary harms of parental substance use on children's educational outcomes: a review. *Journal of Child & Adolescent Trauma*, 15(3), 511–522. <https://doi.org/10.1007/s40653-021-00433-2>
- Loyalka, P., Liu, C., Song, Y., Yi, H., Huang, X., Wei, J., Zhang, L., Shi, Y., Chu, J., & Rozelle, S. (2013). Can information and counseling help students from poor rural areas go to high school? evidence from China. *Journal of Comparative Economics*, 41(4), 1012–1025. <https://doi.org/10.1016/j.jce.2013.06.004>
- Lundberg, S. J., Pollak, R. A., & Wales, T. J. (1997). do husbands and wives pool their resources? evidence from the United Kingdom child benefit. *The Journal of Human Resources*, 32(3), 463. <https://doi.org/10.2307/146179>
- Majeed, S., & Vellakkal, S. (2025). Dry State and fertility: unintended effects of alcohol ban in Bihar. SSRN Scholarly Paper No. 5221412. Social Science Research Network. <https://doi.org/10.2139/ssrn.5221412>
- Mangiavacchi, L., & Piccoli, L. (2018). Parental alcohol consumption and adult children's educational attainment. *Economics & Human Biology*, 28, 132–145. <https://doi.org/10.1016/j.ehb.2017.12.006>
- Marcus, J., & Siedler, T. (2015). Reducing binge drinking? the effect of a ban on late-night off-premise alcohol sales on alcohol-related hospital stays in Germany. *Journal of Public Economics*, 123, 55–77. <https://doi.org/10.1016/j.jpubeco.2014.12.010>
- Markowitz, S., & Grossman, M. (1998). Alcohol regulation and domestic violence towards children. *Contemporary Economic Policy*, 16(3), 309–320. <https://doi.org/10.1111/j.1465-7287.1998.tb00521.x>
- Martin Bassols, N., & Vall Castello, J. (2018). Bar opening hours, alcohol consumption and workplace accidents. *Labour Economics*, 53, 172–181. <https://doi.org/10.1016/j.labeco.2018.04.011>
- McMillan, G. P., & Lapham, S. (2006). Effectiveness of bans and laws in reducing traffic deaths. *American Journal of Public Health*, 96(11), 1944–1948. <https://doi.org/10.2105/AJPH.2005.069153>
- Michaelsen, M. M., & Salardi, P. (2020). Violence, psychological stress and educational performance during the war on drugs in Mexico. *Journal of Development Economics*, 143, 102387. <https://doi.org/10.1016/j.jdeveco.2019.102387>
- Monteiro, J., & Rocha, R. (2017). Drug battles and school achievement: evidence from Rio de Janeiro's Favelas. *The Review of Economics and Statistics*, 99(2), 213–228. https://doi.org/10.1162/REST_a_00628
- Mookerjee, M., Ojha, M., Roy, S., & Yadav, K. (2025). Banning the bottle, shifting the balance: Impact of reduced alcohol consumption on women's agency. Working Paper No. 1662. GLO Discussion Paper. <https://www.econstor.eu/handle/10419/325470>
- Mukherji, R., & Chakraborty, P. (2024). *intoxicant consumption dynamics under alcohol prohibition: evidence from India*. https://icpp.ashoka.edu.in/wp-content/uploads/2025/02/ICPP-Public-Policy-Report_03-1.pdf
- Muralidharan, K. (2024). *Accelerating India's development: a state-led roadmap for effective governance*. Penguin Random House India Private Limited.
- Muralidharan, K., & Prakash, N. (2017). Cycling to school: increasing secondary school enrollment for girls in India. *American Economic Journal: Applied Economics*, 9(3), 321–350. <https://doi.org/10.1257/app.20160004>

- Muralidharan, K., Singh, A., & Ganimian, A. J. (2019). Disrupting education? experimental evidence on technology-aided instruction in India. *American Economic Review*, 109(4), 1426–1460. <https://doi.org/10.1257/aer.20171112>
- Naik, G., Chitre, C., Bhalla, M., & Rajan, J. (2020). Impact of use of technology on student learning outcomes: evidence from a large-scale experiment in India. *World Development*, 127, 104736. <https://doi.org/10.1016/j.worlddev.2019.104736>
- Nakaguma, M. Y., & Restrepo, B. J. (2018). Restricting access to alcohol and public health: evidence from electoral dry laws in Brazil. *Health Economics*, 27(1), 141–156. <https://doi.org/10.1002/hec.3519>
- NFHS-4. (2017). *National Family Health Survey (NFHS-4), 2015-16: India*. International Institute for Science Population (IIPS) and ICF.
- NFHS-5. (2021). *National Family Health Survey (NFHS-5), 2019-21: India*. International Institute for Science Population (IIPS) and ICF.
- Petrusevich, M. (2024). The effects of alcohol sale bans on children: the case of Russia. *Journal of Health Economics*, 97, 102913. <https://doi.org/10.1016/j.jhealeco.2024.102913>
- Pisinger, V. S. C., Møller, S. P., Andersen, S., & Tolstrup, J. S. (2023). Perceived parental alcohol problems and later dropout and grade point average in high school: a register-based follow-up study. *Drug and Alcohol Review*, 42(4), 848–858. <https://doi.org/10.1111/dar.13617>
- Raitasalo, K., Østergaard, J., & Andrade, S. B. (2021). Educational attainment by children with parental alcohol problems in Denmark and Finland. *Nordic Studies on Alcohol and Drugs*, 38(3), 227–242. <https://doi.org/10.1177/1455072520968343>
- Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *The Review of Economic Studies*, 90(5), 2555–2591. <https://doi.org/10.1093/restud/rdad018>
- Reserve Bank of India (2016). Fifth bi-monthly monetary policy statement. Technical report. <https://www.rbi.org.in/commonman/english/Scripts/PressReleases.aspx?Id=2075>.
- Ritchie, H., & Roser, M. (2022). Alcohol Consumption. *Our World in Data*. <https://ourworldindata.org/alcohol-consumption>
- Rossow, I., & Norström, T. (2012). The impact of small changes in bar closing hours on violence. The Norwegian experience from 18 cities. *Addiction*, 107(3), 530–537. <https://doi.org/10.1111/j.1360-0443.2011.03643.x>
- Schoellman, T. (2012). Education quality and development accounting. *The Review of Economic Studies*, 79(1), 388–417. <https://doi.org/10.1093/restud/rdr025>
- Shany, A. (2023). Too Scared for School? Effects of terrorism on students' achievement. *Journal of Human Resources*. <https://doi.org/10.3368/jhr.0721-11759R1>
- Smith, L. C., Ramakrishnan, U., Ndiaye, A., Haddad, L., & Martorell, R. (2003). The importance of women's status for child nutrition in developing countries: International Food Policy Research Institute (IFPRI) Research Report Abstract 131. *Food and Nutrition Bulletin*, 24(3), 287–288. <https://doi.org/10.1177/156482650302400309>
- Smits, J., & Permyer, I. (2019). The subnational human development database. *Scientific Data*, 6(1), 1–15. <https://doi.org/10.1038/sdata.2019.38>
- Wang, H., Sindelar, J. L., & Busch, S. H. (2006). The impact of tobacco expenditure on household consumption patterns in rural China. *Social Science & Medicine*, 62(6), 1414–1426. <https://doi.org/10.1016/j.socscimed.2005.07.032>
- World Health Organization (2025). *Global health observatory data repository*. [https://apps.who.int/gho/data/node.main.A1091?lang=en\(Assessed11/01/2025\)](https://apps.who.int/gho/data/node.main.A1091?lang=en(Assessed11/01/2025)).
- World Health Organization. (2018). *Global status report on alcohol and health 2018*. World Health Organization.
- Young, A. (2019). Channeling Fisher: randomization tests and the statistical insignificance of seemingly significant experimental results*. *The Quarterly Journal of Economics*, 134(2), 557–598. <https://doi.org/10.1093/qje/qjy029>

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